University *of Ljubljana*
Faculty *of Mathematics and Physics*

Department of Physics

Solving Eigenvalue Problems via the Shooting Method

3. Assignment for Model Analysis 2, 2025/26

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1 Introduction

Last week we took a look at how we can use the shooting method to solve Boundary Value Problems (BVPs) which we applied to a mechanical problem where we evaluated wavemodes of a rotating massive rope and classified orbits in the Hénon-Heiles potential. This week we will see that the shooting method is also applicable to a somewhat different class of problems, namely to eigenvalue problems. We will then apply this method to find the eigenenergies and eigenfunctions of the hydrogen atom (its radial part, to be precise) and for solving a *graded index fiber* problem which is described by the Helmholtz equation.

2 Assignment Description

2.1 Hydrogen Atom

The first part of the assignment is to solve the radial part of the Schrödinger equation with a Coulomb potential, which is a well-known problem in quantum mechanics and describes the hydrogen atom. For the hydrogen atom, the equation can be written as:

$$\left[-\frac{\mathrm{d}^2}{\mathrm{d}x^2} - \frac{2}{x} + \frac{l(l+1)}{x^2} - e\right] R(x) = 0\,, \qquad \text{with } R(0) = 0 \text{ and } R(x \to \infty) = 0\,, \tag{1}$$

where $\psi(x) = xR(x)$ is the radial wavefunction, $x = r/a_0$ is the dimensionless radial coordinate, a_0 is the Bohr radius, and $e = E/E_0$ is the dimensionless energy with E_0 being the Rydberg energy. The assignment challenges us to use the Numerov method to solve this problem instead of opting for the standard Runge-Kutta integrator. We can then compare our numerical results to the analytical solutions. For eigenenergies, we expect to find the well-known formula $E_n = -\frac{1}{2n^2}E_0$ where n is the principal quantum number. A state with angular momentum quantum number l will start from the $n = l + 1$ energy level. For the eigenfunctions, we can write the analytical solution as:

$$\psi_{n,l}(x) = \sqrt{\left(\frac{2}{na_0^*}\right)^3 \frac{(n-l-1)!}{2n(n+l)!}}\, e^{-\rho/2} \rho^l L_{n-l-1}^{2l+1}(\rho)\,, \tag{2}$$

where $\rho = 2x/(na_0^*)$ and a_0^* is the reduced Bohr radius. In our case we've set $a_0^* = 1$ for simplicity. The functions $L_{n-l-1}^{2l+1}(\rho)$ are the generalized Laguerre polynomials of degree $n - l - 1$ and order $2l + 1$.

2.2 Graded Index Fiber

The second part of the assignment is to solve the Helmholtz equation for a graded index fiber, which is a type of optical fiber where the refractive index changes gradually with the radius. In our case, the refractive index is given by:

$$n(x) = \begin{cases} 2 - \frac{x^2}{2} & \text{for } x < 1 \\ 1 & \text{for } x \geq 1 \end{cases}\,. \tag{3}$$

The behavior of light in such a fiber can be described by the Helmholtz equation, which is in general given as:

$$\left(\nabla^2 + k^2(\boldsymbol{r})k^2\right)\psi(\boldsymbol{r}) = 0\,. \tag{4}$$

Using the trick of cylindrical symmetry, we can use the ansatz $\psi(x) - (R(x)/\sqrt{x})\,e^{i\lambda z}$ to reduce the problem to a one-dimensional equation for $R(x)$:

$$\left[\frac{\mathrm{d}^2}{\mathrm{d}x^2} + \frac{1}{4x^2} + n^2(x)k^2 - \lambda^2\right] R(x) = 0\,, \qquad \text{with } R(0) = 0 \text{ and } R(x \to \infty) = 0\,. \tag{5}$$

Here, λ is the propagation constant of the mode. The assignment tasks us to find the dispersion relation $\lambda(k)$ for $0.8 \leq k \leq 10$ and to determine which wave numbers k only to single bound state exists (i.e., *single mode fiber*).

3 Solution Overview

In terms of compute stack, absolutely nothing changed since last week. I still used Python along with `numpy`, `scipy` and `matplotlib` for the numerical computations and plotting. The shooting method made use of the root finding capabilities of `scipy.optimize` and the integration was performed using the Numerov method which I implemented myself. Some of the sanity check problems were solved with a PEFRL (Position Extended Forest-Ruth Like) symplectic integrator which I also implemented myself. I wanted to make sure that the shooting method was working independent of the choice of the integrator. For larger parameter sweeps, I again made use of `ray` to distribute tasks across multiple CPU cores.

Last week I complained about the lack of a *sanity check* problem I could solve to verify that my implementation of the shooting method was working correctly. This week I am fortunate that eigenvalue problems are very common in physics and that there are many well-known problems with known analytical solutions that I can use to verify my implementation. As a start, Section 3 shows the results of eigenenergies, eigenfunctions and their absolute errors for the infinite potential well problem aka a particle in a box. This problem was evaluated using the PEFRL integrator with step size of $h = 1/10^4$ and a max energy bound of $E_{\max} = 500$.

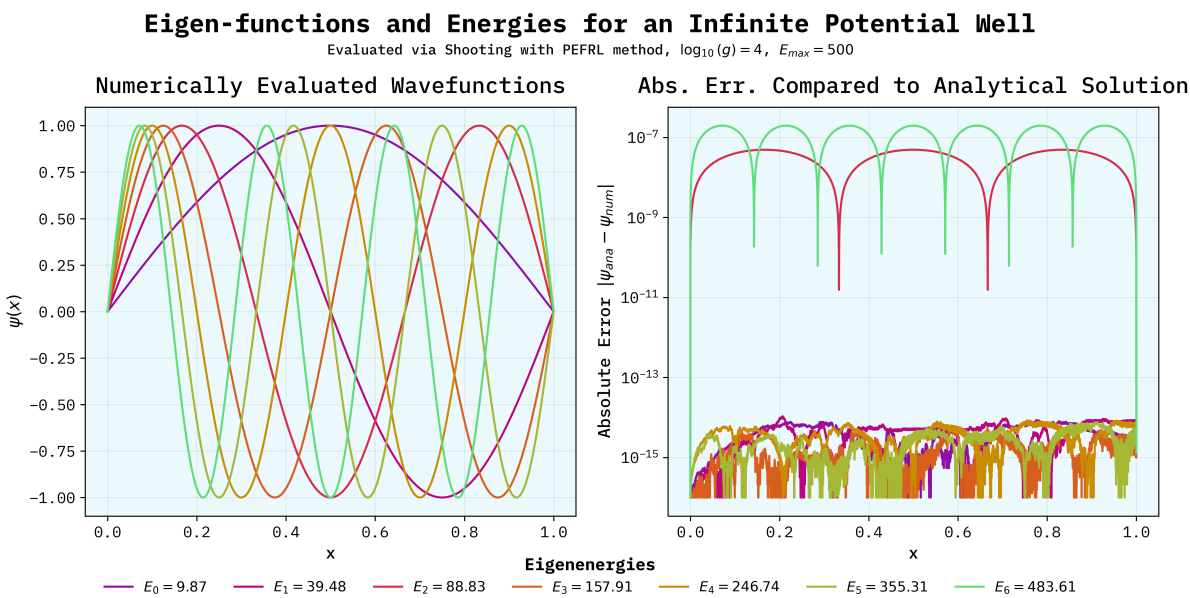

Figure 1: Eigenfunctions and energies for the infinite potential well problem. The left panel shows the numerically obtained eigenfunctions. The right panel shows the absolute error compared to the known analytical solutions.

We see my beloved PEFRL working its magic in the right panel where I've compared the numerical solution to the analytical one. The agreement is excellent, at machine precision for all except E_2 and E_6 which have a larger (still very acceptable) error of around 10^{-7}. To be entirely honest, I'm not sure

what causes these two eigenfunctions to have such a larger error, but even with that, the results are very good, and I am quite confident that my implementation of the shooting method is working. Since we're *shooting for eigenvalues*, it might make sense to visualize this process as well. Section 3 shows the residuals during the root finding process, where the residuals are defined as the value of the wavefunction at the right boundary $x = 1$ for a given energy as we sweep an interval of guesses. The root finding process tries to find the energy values for which this residual is zero, which corresponds to the correct eigenenergies. We can see how neatly the roots are found at the expected energy levels, and how the residuals behave around those roots.

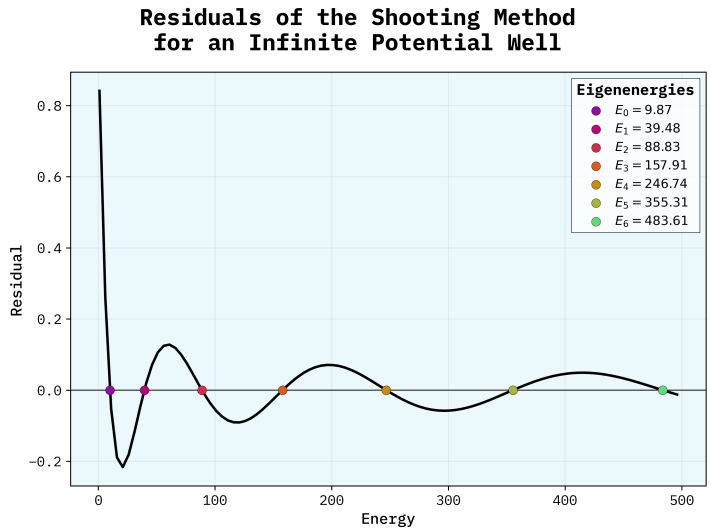

Figure 2: Residuals of the shooting method for the infinite potential well problem. The roots of the residuals correspond to the eigenenergies.

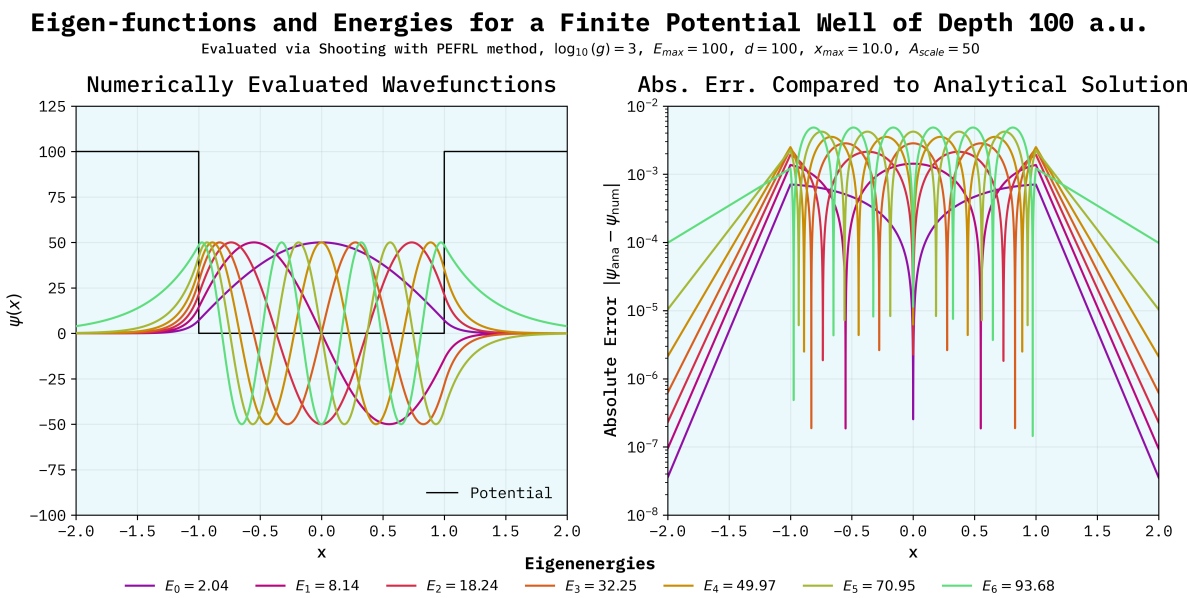

Figure 3: Eigenfunctions and energies for the finite potential well problem. The left panel shows the numerically obtained eigenfunctions. The right panel shows the absolute error compared to the known analytical solutions.

If we want to be **extra** sure that everything is working, we can also look at the related problem of the finite potential well, which also has known analytical solutions. Section 3 shows the results for the finite potential well problem, plotted in the same way as for the infinite potential well. Again evaluated using the PEFRL integrator with step size of $h = 1/10^3$, a depth of $V_0 = 100$ and a max energy bound of

$E_{\max} = 100$. The parameter $x_{\max}$ specifies the extent of the spatial domain over which we solve over. It is larger than what is plotted since due to numerical instability the wavefunctions diverge at large x which we can neatly tuck away. The parameter A_{scale} specifies the scalar by which we multiply the wavefunctions for better visualization.

The results are again pretty good but significantly less accurate than for the infinite potential well. I think it is possible that my implementation of how I solved the analyrical solution for the finite potential well might not be entirely correct, which would explain the larger errors since other than that the eigenenergies are in agreement (relative error below 10^{-4}) with ones obtained via root finding on the transcendental equations. Qualitatively speaking the obtained numerical wavefunctions do look correct, so I still remain confident that things are working. With that I think we should move on to the results for the hydrogen atom and the graded index fiber, which are the main focus of this assignment.

4 Results

4.1 Hydrogen Atom

Section 4.1 shows the results for the hydrogen atom problem. I opted to create a mega compact plot that compares the numerical and analytical solutions for both eigenenergies and eigenfunctions for different values of the angular momentum number $0 \leq l \leq 9$. Since the number of eigenvalues we find varies with l, the y-axis shows the index j of the found eigenenergies for a given l. Eigenenergies were found using the shooting method with the Numerov integrator, a step size of $h = 1/10^5$, a max radial extent of $r_{\max} = 500$ and $n_{\text{scan}} = 5000$ energy guesses.

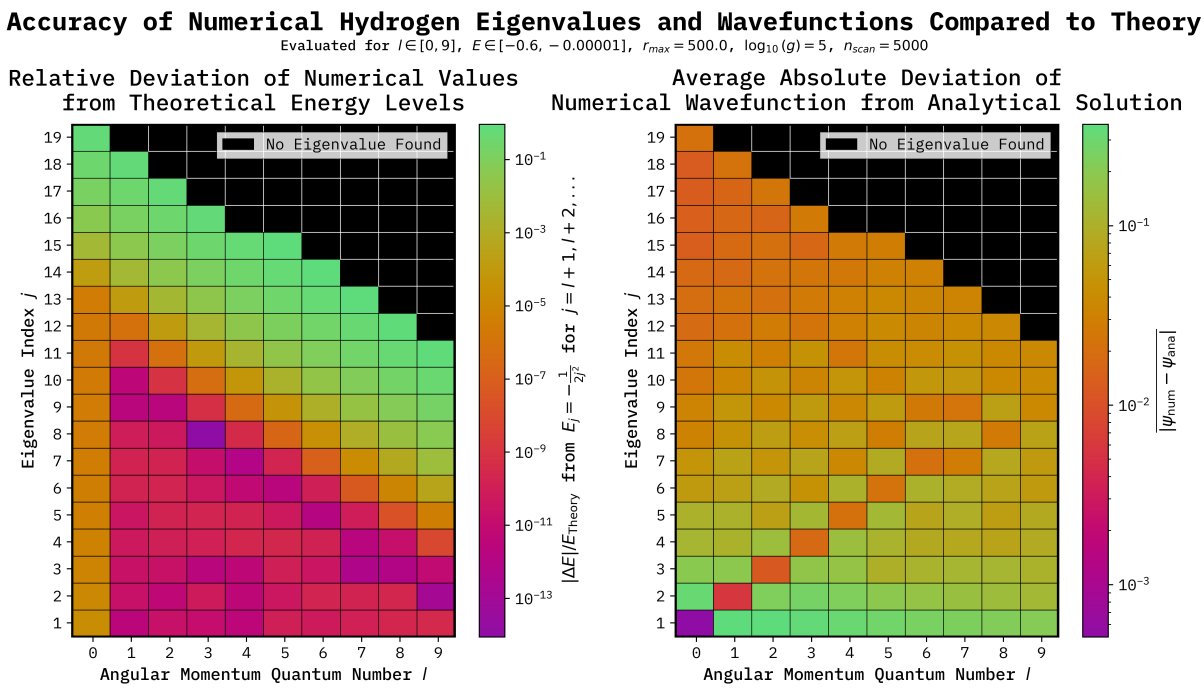

Figure 4: Eigenenergies and eigenfunctions for the hydrogen atom problem. The left panel shows the numerically obtained eigenenergies (dots) compared to the analytical formula (lines). The right panel shows the absolute error of the numerically obtained eigenfunctions compared to the analytical solutions.

I like this plot quite a bit since it serves nicely as a summary of the entire problem. The left panel shows the relative deviation of the numerically obtained eigenenergies and analytical values. Note that we've said that for a given l, the energy levels start from $n = l + 1$, which means that for $l = 0$ we should find eigenenergies starting from $n = 1$, for $l = 1$ we should find eigenenergies starting from $n = 2$ and so on. Overall we find very good agreement with the analytical formula, with relative errors below 10^{-4} for almost all found eigenenergies, except the last couple of eigenenergies for each l which have a larger error of up to 10^{-1}. Most of the lower eigenenergies had a very low relative error of around 10^{-10} or even

lower, which is quite impressive. An interesting asymmetry appears at $l = 0$ however where all values, even ones at low n, have a significantly larger error than for $l > 0$. I think perhaps what might explain this is that out of all of these options, $l = 0$ has the most *singular* behavior at the origin due to the centrifugal term being absent, which appears to make it more difficult to solve numerically. The value at $l = 3, j = 8$ is also interesting since it seems like a numerical fluke where we found an eigenenergy that is abnormaly (given its surrounding values) close to the analytical value.

The right panel shows the absolute error of the numerically obtained eigenfunctions compared to the analytical solutions as described in Equation (2). Here the errors are significantly higher than compared to how accurate the eigenenergies are, with errors around 10^{-3} for the lower eigenfunctions and up to around 10^{-1} for the higher ones. I suspect that this is actually the fault of the integrator since the `get_wavefunction()` method I wrote takes the converged eigenenergy value and performs a final integration to obtain the wavefunction. I have doubts that inaccuracies in the eigenenergy could lead to the accumulation of errors over 10 orders of magnitude. I think I prefer to pin the blame on the Numerov integrator, which sure is theoretically sixth order, but perhaps other integrators could have done a better job. For example, my beloved PEFRL integrator. Perhaps symplicity (or is it symplexity) could have helped here. Unfortunately, due to limited time I was not able to test this hypothesis. There is another interesting pattern we can spot in the right panel. A diagonal line forms in the plot where the error is significantly lower than the surrounding values. This line starts at $l = 0, j = 1$ and continues upwards and to the right for one value until $l = 7$ where I'm not sure if I can comfortably claim that a trend is still visible. The beginning of this line starts at the lowest energy level for $l = 0$ which is the $n = 1$ state and has by far the lowest error of all the states. The next point on the line is at $l = 1, j = 2$ which corresponds to the $n = 3$ state, which also has a very low error and so forth. If I understand my indexing notation correctly it would seem that states $j = l + 1$ for a given l have a significantly lower error. That corresponds to the states with $n = 2l + 1$, which which are all odd n states. Interestingly, I think one can imagine the next diagonal for $j = l + 3$ which is barely visible. These would also correspond to odd n states. I think its possible that either the odd states are easier to compute or that the analytical solution and the way I evaluate it via `scipy.special.genlaguerre()` is different for odd and even n states. Not sure... perhaps I'm looking too deeply into something that is just noise.

5 Conclusion and Comments

One oversight that I have regrets over is that some of the plots used a 0-indexed n for energy levels instead of the more expected 1-indexed n.